Comparative analysis of income distributions in the European Union and the United States

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Abstract

We prove that the most rafined approach – our extension of the Yakovenko et al. model – is a universal in the sense that it well describes both household incomes in the European Union and the individual incomes in the United States for all income society classes. This prove was based on our comparative study of various kinds of incomes. The study consitutes a basis for the finding of an impact of the recent world-wide financial crisis on the volatility of various temporary Pareto exponents and on other parameters of the model.

Keyword: Household income, Individual income, Yakovenko model, Pareto exponent, Complementary cumulative distributon function

1 Universal formula

One of the major trends in socio- and econophysics having a long tradition is the study of income and wealth redistribution in society and the analysis of social inequalities. Several models, trying to explain the microscopic mechanisms of income dynamics of individuals or households, were proposed [1, 2] (and refs. therein). However, none of them give an analytical description of the incomes for all society classes by a single formula based on a unified formalism. Recently [2], we extended the Yakovenko et al. model providing, indeed, such a universal formula. Here we show that this formula, containing a low number of free parameters, well reproduces the empirical complementary cumulative distribution functions (CCDFs) both for the European Union (EU) and for the United States (US). Notably, the CCDF is the main statistical tool commonly used in this context. This is the probability that independent stochastic variable takes value larger than some fixed one.

As for the Yakovenko et al. model, the coexistence of additive and multiplicative stochastic processes is also allowed for our extended Yakovenko et al. model. That is, we assume that household or individual income is determined by: (i) wages and salaries and/or (ii) profits which go to households or individuals mainly through investments and capital gains. Furthermore, for the extended Yakovenko et al. model we assume that the formalism of the income change is the same for entire society but its particular dynamics distinguishes the range of the high-income society class from that of the others (see [2] for details). Finally, the equilibrium probability distribution function we derived in the form [2],

$$P_{\rm eq}(m) \propto \begin{cases} \frac{\exp(-(m_0/T)\arctan(m/m_0))}{[1+(m/m_0)^2]^{(\alpha+1)/2}}, & m < m_1 \\ \frac{\exp(-(m_0/T_1)\arctan(m/m_0))}{[1+(m/m_0)^2]^{(\alpha_1+1)/2}}, & m \geq m_1 \end{cases}$$

where parameter T can be interpreted as an average income per household or individual within the low- and medium-income society classes, while parameter T_1 can be interpreted as an average income per household or individual within the high-income society class. The shape parameter α is the Pareto exponent, describing the income inequality within the medium-income society class, and the shape parameter α_1 is the Pareto exponent, describing the income inequality within the high-income society class. Parameter m_0 is a crossover (border) income between the low- and medium-income society classes, and m_1 is a crossover income between the medium- and high-income society classes. The complementary cumulative distribution function considered below is, indeed, the integrated quantity of above given distribution function.

2 Agreement with empirical data

In Fig. 1 we compare the predictons (shown, for instance, for 2007) based on the extended Yakovenko et al. formula with: (i) empirical complementary cumulative distribution functions of the annual adjusted gross income of individuals in the US and (ii) empirical complementary cumulative distribution functions of the annual total gross income of households in the EU. The observed good agreements have significant consequences.

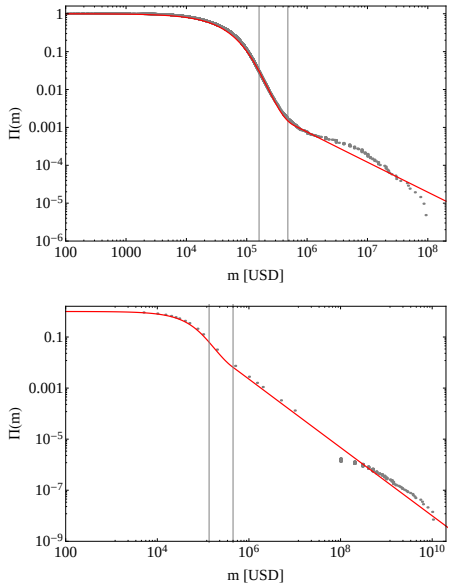


Figure1: The comparison for year 2007 of the complementary cumulative distribution functions based on our extended Yakovenko formula (solid lines) with the EU household income empirical data set (dots; upper plot) and the US individual income data set (dots; lower plot) [3]. The first and the second vertical lines are placed at m_0 and m_1, respectively. We found for the years 2005-2010: (i) for EU $174\,495 \leq m_0 \leq 208\,116$, while (ii) for US $135\,000 \leq m_0 \leq 150\,000$.

Apparently, the middle-income society class is strongly reduced (almost absent) in US in comparison with EU. This result is persistent, i.e. valid for each considered year (in our case from 2005 to 2010). Besides, the border, m_0, between the low- and middle-income society classes increased at the very beginning of the recent world-wide financial crisis – by about 10% at 2006 in US and by about 20% at 2007 in EU – increasing, thereby, the ranges of the corresponding low-income society classes. Perhaps, these could be identified as early-warning signals preceding the crisis – however, to say somethig more definite, it requires a much more systematic study. Furthermore, the drastic decrease of the border, m_1, between the middle- and high-income society classes – by about 65% at 2009 in EU and almost by the same amount at 2006 in US (see Table 1) – increases the relative range of the high-income society classes. The former border decrease directly relates to the most striking observation that the high-income society class in EU vanishes at 2009 as then both shape exponents (α and α_1) are almost equal (up to about 10% accuracy, see Table 2). Again, it is worthy of a much more systematic study.

Table 1: Parameters T and $T_1(= m_1)$ obtained (in US dollars) for the years 2005-2010 from the fit of the CCDF based on our extended Yakovenko et al. formula to the corresponding empirical ones concerning the annual: (i) gross income of individuals in the US and (ii) total gross income of households in the EU. The error bars of the parameters do not exceed 18%.

	European Union		**United States**	
Year	T	T_1	T	T_1
2005	46 278	552 770	45 520	380 000
2006	43 985	529 006	47 220	350 000
2007	48 127	624 350	48 430	450 000
2008	55 257	654 355	48 740	460 000
2009	47 448	371 890	48 050	500 000
2010	51 574	610 749	48 680	420 000

Table 2: Parameters m_0, α and α_1 obtained for the years 2005-2010 from the comparison of the theoretical CCDF based on the extended Yakovenko et al. formula with: (i) empirical CCDFs of the annual total gross income of households in the EU, and (ii) empirical CCDFs of the annual adjusted gross income of individuals in the US. The error bars of exponents α and α_1 do not exceed 4%.

	European Union		**United States**	
Year	α	α_1	α	α_1
2005	2.907	0.795	1.93	1.354
2006	2.892	0.86	1.88	1.346
2007	2.735	0.79	1.83	1.336
2008	2.965	0.890	1.85	1.381
2009	2.974	2.608	1.90	1.451
2010	3.153	0.77	1.86	1.395

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